BEFORE YOU READ

Properties in matched pairs

The properties of arithmetic come in matched pairs for addition and multiplication, with the distributive property bridging the two.

- L0 **Properties of operations**
  - L1 **Addition**
    - L2 **Commutative** $a + b = b + a$
    - L2 **Associative** $(a + b) + c = a + (b + c)$
    - L2 **Identity** 0
    - L2 **Inverse** $-a$
  - L1 **Multiplication**
    - L2 **Commutative** $a \cdot b = b \cdot a$
    - L2 **Associative** $(ab)c = a(bc)$
    - L2 **Identity** 1
    - L2 **Inverse** requires $a \neq 0$ $1/a$
    - L2 **Zero property** no additive analog $a \cdot 0 = 0$
  - L1 **Bridge**
    - L2 **Distributive** $a(b + c) = ab + ac$

TAKEAWAY

One property breaks the symmetry

The zero property, $a \cdot 0 = 0$, has no counterpart for addition.